

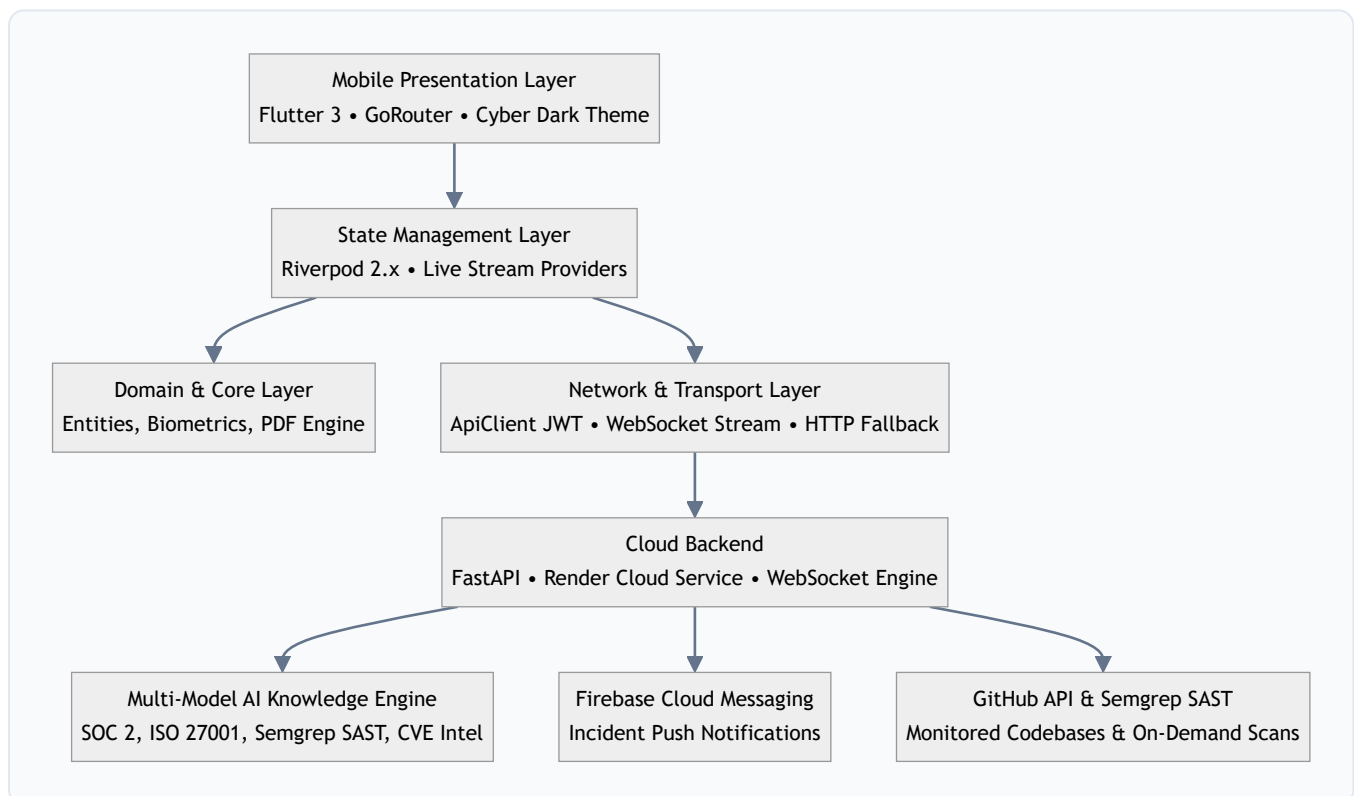
SecurePulse

 FLUTTER 3.X DART 3.X FASTAPI CLOUD BACKEND FIREBASE FCM PUSH

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SecurePulse is a next-generation enterprise cybersecurity and SOC mobile operations platform built for security analysts, incident responders, and CISOs. Designed with a high-contrast dark cyberpunk aesthetic, hardware-backed biometric security, live WebSocket threat streaming, dynamic AI security assistance, real GitHub SAST repository scans, and one-tap executive PDF compliance report exports.

Architecture & Tech Stack



Core Technologies

- **Mobile Frontend:** Flutter 3 (Dart 3), Riverpod State Management, GoRouter, `flutter_animate`, `fl_chart`.

- **Cloud Backend:** Python 3.11, FastAPI, Uvicorn, WebSockets, JWT Authentication, Render Cloud.
- **Push Notification Pipeline:** Google Firebase Cloud Messaging (FCM) & `firebase_admin` SDK.
- **Document Engine:** Pure vector `pdf` engine with native `share_plus` and `path_provider` export.
- **Hardware Security:** `local_auth` biometric authentication (Fingerprint / Face ID).

Key Platform Capabilities

1. Executive Security Operations Dashboard

- Real-time **Security Posture Score** (94/100, Grade **A+**) with animated gauge rings and vulnerability charts.
- Live system status monitors for Cloud API, WebSockets, GitHub integration, and AI engine.
- Instant access to quick actions: One-tap SAST scans, SOC incident triage, AI chat, and compliance reports.

2. Real-Time WebSocket SOC Threat Streaming

- Subscribes to live WebSocket telemetry (`wss://.../ws/alerts`) for instant incident detection.
- Resilient architecture with **automatic HTTP long-polling fallback** if network proxies block raw sockets.
- Incident severity badges (Critical, High, Medium, Low), attack vector classifications, and IP origin telemetry.

3. GitHub Repository Integration & Live SAST Scans

- Monitored codebases include `NATTOMR/secureguard-mobile` and `NATTOMR/secureguard-backend`.
- One-tap "**Trigger Immediate SAST Scan**" dispatches real-time Semgrep & Secret scanning across cloud repositories.
- Live telemetry sync updates repository health grades and findings counts on your phone.

4. Dynamic AI Security Copilot (`SecurePulse AI Agent`)

- Expert cybersecurity knowledge engine for **SOC 2 Type II**, **ISO/IEC 27001:2022**, and zero-day threat patches.
- Generates tailored remediation patches, `iptables` / `ufw` / Cloudflare firewall rules, and custom Semgrep YAML rules.

5. Executive PDF Compliance Report Export

- One-tap generation of official audit PDFs for:
- **SOC 2 Type II** (Trust Services Criteria CC6.1 - CC8.1)
- **ISO/IEC 27001:2022** (Annex A.5 - A.8)
- **PCI-DSS v4.0 & HIPAA Security Rule**

- Includes cryptographic SHA256 audit verification stamps and direct export via the native Android Share Sheet (Email, Slack, WhatsApp, Google Drive).

6. 🎨 Custom Cyber Shield Launcher & Splash Branding

- Custom dark obsidian metallic app icon with a **neon cyan holographic cyber defense shield and circuit lock core**.
- Android adaptive launcher icons (`mdpi` , `hdpi` , `xhdpi` , `xxhdpi` , `xxxhdpi`) and shimmering animated splash screen.

📁 Project Directory Structure

```
secureguard-mobile/
├── android/                # Android native configuration & Gradle scripts
│   ├── app/
│   │   ├── google-services.json # Firebase configuration
│   │   └── src/main/res/        # Custom adaptive cyber launcher icons
├── assets/
│   └── images/                # High-resolution app icon & branding assets
├── lib/
│   ├── app.dart              # Root MaterialApp & theme configuration
│   ├── main.dart             # App initialization (Firebase, Biometrics, Logging)
│   ├── core/                 # Core infrastructure layer
│   │   ├── config/           # Environment & backend URL configurations
│   │   ├── constants/        # AppColors, AppStrings, AppAssets
│   │   ├── network/          # ApiClient (JWT auto-auth), WebSocketService (with fallback)
│   │   ├── router/           # GoRouter with bottom navigation shell
│   │   ├── services/         # BiometricService, NotificationService, ReportPdfService
│   │   ├── storage/          # SecureStorageService (JWT token storage)
│   │   ├── theme/            # Cyber Dark theme tokens & Google Fonts
│   │   └── widgets/          # SGCard, SGButton, SGAppBar, SGStatisticCard, etc.
│   ├── features/             # Feature-First modular architecture
│   │   ├── ai/               # SecurePulse AI Agent chat screen & repository
│   │   ├── alerts/           # SOC incident alerts & incident detail investigation
│   │   ├── auth/             # Login screen with GitHub, Email, Biometric auth
│   │   ├── dashboard/        # Executive security dashboard & charts
│   │   ├── findings/         # Vulnerability findings & severity filters
│   │   ├── profile/          # Analyst profile & credentials
│   │   ├── reports/          # Executive compliance reports & PDF export
│   │   ├── repositories/     # Monitored codebases & SAST scan triggers
│   │   ├── scans/            # Scan history & detailed log viewer
│   │   ├── settings/         # Environment switching & platform settings
│   │   └── splash/           # Branded animated splash screen
│   └── providers/            # Riverpod providers & live stream notifiers
└── test/                     # Unit & integration test suite (15/15 passing)
```

Complete Documentation Suite

Comprehensive technical, architectural, and operational documentation is available in the [docs/](#) directory:

Document	Description
 Project Technical Report	Executive project dossier, problem matrix, test coverage, and security audit.
 Architecture Mindmap	Visual Mermaid system architecture, dual mode dataflows, and navigation graphs.
 Operations Walkthrough	End-to-end screen workflows, triage steps, and developer user manual.
 Product Roadmap	Strategic evolution phases, milestone timelines, and feature deliverables.
 API Integration Guide	FastAPI endpoint contracts, WebSocket streaming, and deployment guides.

Getting Started

Prerequisites

- [Flutter SDK](#) ≥ 3.0.0
- [Dart SDK](#) ≥ 3.0.0
- Android Studio / Android SDK Platform 34+

Setup & Run

```
# 1. Clone repository
git clone https://github.com/NATTOMR/secureguard-mobile.git
cd secureguard-mobile

# 2. Fetch dependencies
flutter pub get

# 3. Run test suite
flutter test

# 4. Run on connected device or emulator
flutter run
```

Building Release APK

```
flutter build apk --release
```

The compiled APK will be generated at `build/app/outputs/flutter-apk/app-release.apk` (and mirrored at `securepulse-release.apk` in the root folder).

Backend Cloud Endpoints

The mobile app connects to the live **SecurePulse Cloud Service** hosted on Render:

Endpoint	Method	Description
<code>/v1/auth/login</code>	POST	Analyst JWT authentication
<code>/v1/dashboard</code>	GET	Overall security posture score & metrics
<code>/v1/alerts</code>	GET	Active SOC security incidents
<code>/ws/alerts</code>	WSS	Real-time WebSocket threat stream
<code>/v1/repositories</code>	GET	Monitored codebases telemetry
<code>/v1/repositories/{id}/scan</code>	POST	Trigger on-demand SAST security scan
<code>/v1/ai/chat</code>	POST	Live AI Security Copilot query
<code>/v1/reports</code>	GET	Executive compliance report logs

Security & Privacy

- **Zero-Trust Client Sessions:** Hardware-backed biometric authentication with auto-expiring JWT bearer tokens.
- **Encrypted Transmission:** All cloud traffic enforced over **TLS 1.3 / WSS** with strict CORS and rate limiting.
- **Push Protection:** Automated secret scanning to prevent credential leakage.

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